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SLAC National Laboratory

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2575 Sand Hill Road
Menlo Park, CA 94025

Job Application for: Beamline Engineer - SSRL Materials Science Division

Dear Hiring Manager,

About Me

I am excited to apply for the Beamline Engineer position within SSRL's Materials Science Division. I currently work as a Staff Engineer at SLAC, where I focus on restoring, optimizing, and supporting accelerator and diagnostic systems. This role strongly aligns with my interest in hands-on engineering, experimental systems, and user support—areas where I have built a solid foundation and continue to grow.

At SLAC, I have led the recommissioning of critical beamline systems, including the FACET positron beamline and a dormant bunch length diagnostic. These efforts required full ownership across vacuum systems, electrical infrastructure, cooling systems, and controls, with a focus on bringing complex, inactive systems back into reliable operation. In parallel, I have developed MATLAB-based tools for data acquisition and visualization to better understand and optimize beam performance, supporting both operations and experimental work.

Prior to SLAC, I worked at Fermilab as part of the US HL-LHC Accelerator Upgrade Project, where I focused on superconducting magnet R&D, manufacturing, and electrical quality assurance. I worked directly on the fabrication, testing, and validation of Nb_3Sn superconducting magnet systems designed to operate under extreme thermal conditions, from cryogenic environments to high-temperature processing. I led electrical QA efforts, supported failure analysis, and collaborated closely with technicians and scientists to improve fabrication reliability and performance. This experience gave me a strong understanding of how complex systems are built, how they fail, and how to design and test them for robustness.

Why I Fit

Across both roles, I have worked in highly collaborative environments, interfacing regularly with physicists, engineers, technicians, and operations teams. I am comfortable taking ownership of systems, troubleshooting issues in real time, and supporting experimental needs in fast-paced settings. I am particularly drawn to this role at SSRL because of the opportunity to work more directly with beamline instrumentation and user-facing experimental systems while continuing to improve system reliability and performance.

While my background has been in accelerator beamlines, I am eager to apply that systems-level engineering experience to hard X-ray beamlines and experimental stations. I am confident that my combination of hands-on engineering, R&D experience, and operational support would allow me to contribute effectively to SSRL from day one.

I would welcome the opportunity to discuss how my background aligns with the needs of your team.

Sincerely,

Marcellus E. Parker